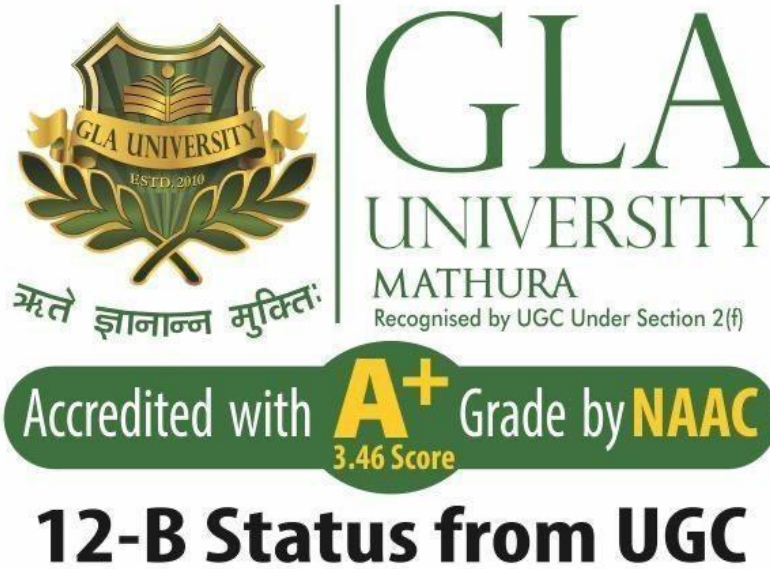


B.Tech CSE (III YEAR - V SEM) (2026-27)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

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(Uttar Pradesh) India

Project Title : IndustrialGuard AI: AI-Powered Proactive Industrial Safety Monitoring & Risk Intelligence Platform

Team Leader : Anjali Kumari UR: 2415500072

Team Member : Abhay Singh UR: 2415500008

Mentor Name : Sir Govind Gupta

Signature : _____

Govind
03/09/2026

Version: v1.0 Date: 30 Aug 2026

Project Synopsis

IndustrialGuard AI: AI-Powered Proactive Industrial Safety Monitoring & Risk Intelligence Platform

Field	Details
Project title	IndustrialGuard AI: AI-Powered Proactive Industrial Safety Monitoring & Risk Intelligence Platform
Team Number	125
Institute / Course	GLA University, Mini-Project
Version	v1.0
Date	30 Aug 2026

1. Overview

Problem statement: Industrial safety monitoring is often fragmented across manual inspections, CCTV feeds, environmental sensors and incident records. Hazards may be detected late, while historical data is difficult to convert into leading indicators and preventive action.

Goal: Build a modular AI-powered safety intelligence platform that combines environmental sensor signals and computer vision to detect hazards, estimate risk, trigger alerts, and maintain incident intelligence for industrial sites.

Non-goals: Autonomous control of industrial machinery, safety certification, medical diagnosis, employment decisions, biometric identification, and fully autonomous emergency response are out of scope for v1.

Value proposition: A lightweight, explainable safety intelligence layer designed for small and medium industrial facilities that can start with simulated or low-cost sensors and existing camera feeds, then scale to edge/IoT deployments.

2. Scope and Control

2.1 In-scope

- User and site setup with role-based access
- Environmental sensor ingestion: temperature, humidity, gas and smoke values
- Real-time sensor dashboard and trend visualisation
- Machine-learning risk classification: SAFE / WARNING / CRITICAL
- Computer-vision MVP for selected PPE scenarios such as helmet/vest detection
- Multi-signal risk fusion and incident creation
- Real-time alerts and incident history
- Analytics for risk trends, hotspots and resolution status

2.2 Out-of-scope

- Direct PLC or machine emergency-stop control in v1
- Full autonomous industrial control
- Biometric identification or facial recognition
- Large-scale predictive maintenance
- Mobile-native applications
- Multi-site enterprise orchestration beyond prototype scope

2.3 Assumptions

- A representative sensor dataset can be obtained or simulated for model development.
- Camera footage or public/academic datasets can be used for the selected PPE scenario.
- Prototype users are safety supervisors or plant administrators with basic dashboard familiarity.
- Connectivity is available for the software prototype; the optional physical demo may use Wi-Fi via ESP32.

2.4 Constraints

- Academic mini-project timeline and two-member development team
- Limited access to production industrial sensor data
- Compute limits for computer-vision experimentation
- Need to validate the system without making unsafe real-world control decisions

2.5 Dependencies

- Python and Scikit-learn environment
- Computer-vision framework/model and representative PPE dataset
- FastAPI backend and WebSocket support
- React/Tailwind dashboard
- SQLite for prototype persistence; PostgreSQL-ready schema for scale-up
- Optional ESP32 + low-cost sensors for physical demonstration

2.6 Acceptance criteria and sign-off

- Given valid sensor readings when the data enters the API then the values are validated, stored and displayed on the dashboard.
- Given a trained risk model when the required features are available then the system returns a safety class and risk score within the defined prototype latency target.
- Given a detected high-risk visual or sensor event when the risk fusion threshold is crossed then an incident and alert are created.
- Given an authorized supervisor when an incident is reviewed then its status, notes and resolution are recorded.

Stakeholder	Role	Decision area	Signature/Approval	Date
Mr. Govind Gupta	Mentor	Scope, architecture, final acceptance	Approved	03/09/2026
Anjali Kumari	Team Lead	Technical readiness	Approved	01/09/2026
Abhay Singh	Product / QA Lead	Documentation and validation readiness	Approved	01/09/2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Abhay Singh	Anjali Kumari	Mentor	Team
Design	Anjali Kumari & Abhay Singh	Anjali Kumari	Mentor	Team
Implementation	Anjali Kumari	Anjali Kumari	Mentor	Team
Testing	Abhay Singh	Anjali Kumari	Mentor	Team
Release	Abhay Singh	Anjali Kumari	Mentor	Department

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Anjali Kumari	Tech Lead & ML/CV Backend	Architecture, ML pipeline, CV integration, FastAPI, security	Python, FastAPI, Scikit-learn, OpenCV, SQL	10 hrs/wk	anjali.kumari_cs.aiml24@gla.ac.in
Abhay Singh	Product, Frontend & QA/Docs	Research, requirements, React UI, test plan, documentation, demos	React, UI/UX, testing, technical writing	10 hrs/wk	abhay.singh_cs.aiml24@gla.ac.in

5. Week-wise Plan and Assignments

Week	Milestone	Anjali (Tech/ML)	Abhay (Product/UI/QA)	Deliverables	Status
1	Requirements & research	Architecture notes	Problem validation + competitors	Synopsis draft	Planned
2	Data & system design	Data schema + ML plan	UX flows + acceptance criteria	SRS/spec v1	Planned
3	Backend + baseline ML	FastAPI + Random Forest baseline	Dashboard shell + test data	MVP data path	Planned
4	Computer vision	PPE detector prototype	UI event views + QA	CV feature demo	Planned
5	Risk fusion	Risk scoring logic	Incident workflow	Integrated MVP	Planned
6	Testing & hardening	Model tuning + API tests	E2E + documentation	Test report v1	Planned
7	Controlled prototype demonstration	Deployment + performance	Demo script + bug fixes	Demonstration build	Planned
8	Release & presentation	Final technical review	Final synopsis + slides	Final synopsis, prototype and presentation	Planned

6. Users and UX

6.1 Personas

- **Safety Officer:** Needs a clear view of active hazards, incident severity, trends and actions requiring follow-up.
- **Shift Supervisor:** Needs fast alerts and concise incident context to coordinate immediate local response.
- **Plant Manager:** Needs aggregated trends, hotspots, resolution metrics and evidence of continuous safety improvement.

6.2 Top User Journeys



6.3 User Stories

- As a safety officer, I want to see live sensor and AI risk status so that I can identify unsafe conditions quickly.
- As a supervisor, I want alerts to include severity and evidence so that I can decide what action is appropriate.
- As a plant manager, I want weekly risk trends and hotspots so that I can prioritize preventive measures.

6.4 Accessibility & Localization

- High-contrast safety states and text labels
- Color is not the only indicator; severity also includes text and icons
- Responsive web dashboard for desktop/tablet use
- English language support in v1; localization can be added later

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Observed gap vs. our target	Our differentiator
Intenseye	Industrial EHS AI	Manufacturing / enterprise	50+ safety indicators; CV; EHS workflows; integrations	Public pricing not listed on cited product pages	Broad enterprise capability; industrial integrations	Higher deployment breadth and complexity than our MVP target	Modular, lower-complexity SME-oriented prototype
Protex AI	Site intelligence for safety + operations	Industrial, logistics, warehousing	PPE, area control, vehicle control, behavioural safety, analytics	Public pricing not listed on cited pages	Uses existing cameras; safety + operations analytics	Enterprise-oriented product scope	Affordable modular sensor + CV architecture
Voxel	Industrial physical AI	Industrial operations / safety	Real-time CV, risk detection, workflows, reporting	Public pricing not listed on cited pages	End-to-end detection-to-resolution workflow	Broad industrial deployment focus	Student-built transparent architecture and low-cost MVP
viAct	Industrial AI computer vision	Construction / manufacturing / industrial	Safety, PPE, operational monitoring, AI vision	Public pricing not listed on cited page	Large scenario library and video analytics	Broader scenario coverage than our initial MVP	Focused, explainable MVP for selected hazards
Visionify	AI video analytics for safety	Factories / workplaces	PPE and safety monitoring using cameras	Public pricing not listed on cited page	Lightweight camera-based safety use cases	Primarily vision-centric	Sensor + vision risk fusion in one prototype

Market sources: Intenseye Core AI / Industrial Integrations; Protex AI Safety / Privacy; Voxel Physical AI for Industrial Operations; viAct manufacturing safety materials; Visionify product materials. See References.

7.2 Positioning

IndustrialGuard AI focuses on affordable, modular, explainable safety intelligence for small and medium industrial facilities. The design intentionally avoids claiming parity with mature enterprise platforms; instead it provides a buildable proof of concept with a clear path to scale.

Measurable Delta (MVP targets)

- Sensor-to-dashboard update: target ≤ 2 seconds under prototype conditions.
- Alert creation after a confirmed high-risk event: target ≤ 3 seconds.
- Backend API: target p95 latency ≤ 1000 ms for standard dashboard/API requests.
- Model reporting: publish accuracy, precision, recall and F1 rather than relying on accuracy alone.
- Operational traceability: every generated incident has timestamp, source signal, severity and resolution state.

8. Objectives and Success Metrics

Objective	Area	Target	KPI
O1	Data ingestion	Validate and store sensor data reliably by MVP review	$\geq 95\%$ valid records accepted without manual correction
O2	Risk classification	Build baseline model for SAFE/WARNING/CRITICAL states	Publish F1 score and confusion matrix on held-out test set
O3	Vision safety	Detect selected PPE conditions on representative test images/video	Report precision, recall and inference behaviour
O4	Real-time monitoring	Connect prediction pipeline to dashboard and alerts	End-to-end event visible in ≤ 3 s target
O5	Reliability	Document test evidence and address critical issues before submission	No unresolved critical defects at submission

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Sensor Monitoring	Collect, validate and display temperature, humidity, gas and smoke signals.	Must	ESP32 or simulated data; FastAPI	GIVEN a valid reading WHEN it reaches the API THEN it is validated and appears on the dashboard.
Risk Classification	Classify the current safety state using engineered sensor features and a Random Forest baseline.	Must	Pandas, Scikit-learn, stored model	GIVEN required features WHEN prediction runs THEN the system returns class + score.
PPE Detection	Detect selected PPE states such as helmet or safety vest presence in sample images/video.	Must	OpenCV + CV model + dataset	GIVEN an input frame WHEN a target PPE class is present/absent THEN the event is recorded with confidence.
Risk Fusion	Combine sensor and vision signals into an overall severity score.	Must	Risk rules / weighted fusion	GIVEN multiple signals WHEN thresholds are crossed THEN a single incident severity is generated.
Alerts	Generate dashboard/browser and optional email/buzzer alerts.	Should	WebSocket + notification layer	GIVEN a critical event WHEN alert policy matches THEN an alert is emitted and logged.
Incident Management	Create, assign, review and close safety incidents.	Must	SQLite/PostgreSQL-ready schema	GIVEN an incident WHEN a supervisor updates it THEN state and notes are persisted.
Analytics	Show trends, hotspots and resolution status over time.	Could	Recharts + aggregated queries	GIVEN historical incidents WHEN analytics loads THEN relevant metrics are displayed.

9.1 Feature Prioritisation and MVP Boundary

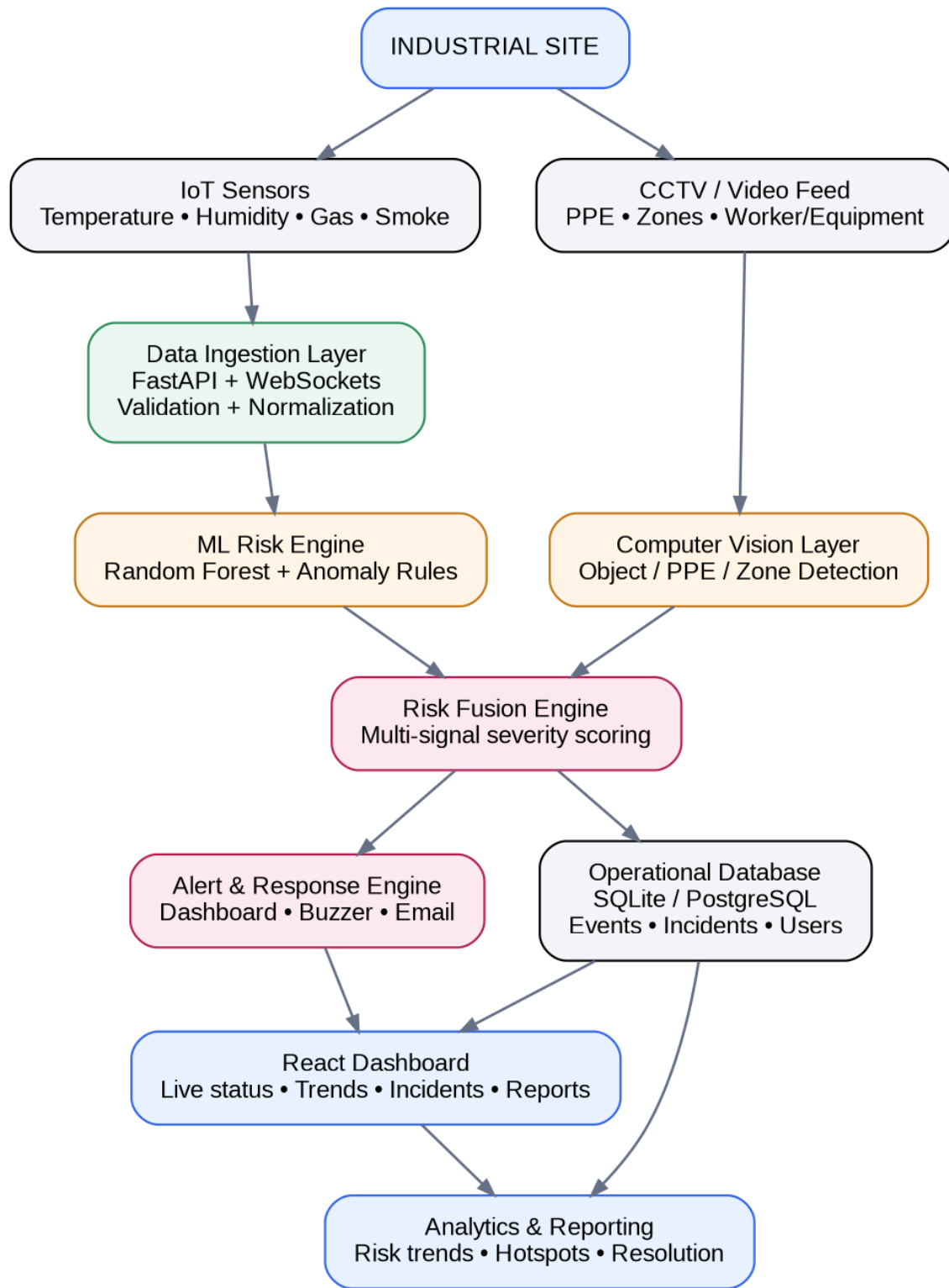
Priority	Features	Reason
Must	Sensor monitoring, risk classification, risk fusion, alerts, incidents, dashboard	These form the minimum end-to-end safety intelligence loop.
Should	PPE detection, email notifications, site/zone tagging	Adds strong product value without blocking the core loop.
Could	Advanced anomaly detection, predictive trends, multi-site analytics	Useful extensions after the MVP is stable.
Won't in v1	PLC control, facial recognition, autonomous machine shutdown, mobile native app	Safety-critical or high-complexity capabilities outside the academic prototype scope.

9.2 Core Safety States

State	Interpretation	System action
SAFE	Signals remain within configured operating range and no high-confidence visual hazard is present.	Continue monitoring; no escalation.
WARNING	One or more leading indicators show elevated but non-critical risk.	Create warning event; notify dashboard; supervisor review as needed.
CRITICAL	Combined risk score or safety event crosses the critical threshold.	Create incident; issue immediate alert; record evidence and timestamp.

Thresholds are configuration values for the prototype and must be validated against the chosen dataset; they are not presented as medical, legal or industrial safety certification limits.

10. Architecture



Architecture: proposed IndustrialGuard AI MVP design. Sensor and computer-vision streams converge in a risk-fusion layer before alerting, storage and analytics.

10.1 High-Level Architecture

The system follows a modular architecture designed to support real-time sensor ingestion, computer-vision events, explainable risk scoring and incident workflows.

Clients

- Web-based React single-page application provides live status, incident review, analytics and configuration screens.
- Optional physical demo uses ESP32 sensors to emulate or provide live environmental signals.

Backend Services

- Ingestion Service: validates sensor and event payloads.
- Risk Service: loads the trained ML model and computes risk class/score.
- Vision Event Service: accepts CV detections and normalises event metadata.
- Incident Service: creates, updates and closes incidents.
- Alert Service: dispatches dashboard/WebSocket and optional notification events.

AI/ML Services

- Random Forest risk classifier for tabular sensor features.
- Computer-vision detection model for the selected PPE scenario.
- Optional anomaly rules for sudden sensor deviations.
- Risk fusion layer combining heterogeneous signals.

Data Stores

- SQLite for prototype; PostgreSQL-ready relational design for scale-up.
- Event metadata, sensor readings, predictions, incidents and audit logs are persisted.

Integrations

- ESP32/HTTP or WebSocket for optional IoT demo.
- Sample video or IP camera feed for computer-vision input.
- SMTP/browser notifications are optional extensions.

10.2 Planned API & Communication Layer

- Planned API responsibilities: sensor ingestion, risk prediction, computer-vision events, incident management and analytics queries.
- REST endpoints will be defined during implementation after the data model and user flows are validated.
- WebSocket communication is planned for near-real-time dashboard updates and alert delivery.
- Authentication and role checks will protect write operations and incident evidence.

10.3 Configuration and Secrets

- Environment variables managed through local .env files and deployment secrets.
- API keys, SMTP credentials and JWT secrets are never committed to Git.
- Sensitive values are excluded from logs and screenshots.
- Prototype uses role-based access; service-to-service secrets are configurable for future deployment.

11. Data Design

11.1 Core Entities

- User
- Site
- Sensor
- SensorReading
- VisionEvent
- RiskAssessment
- Incident
- Alert
- AuditLog

11.2 Proposed Data Model

- Core entities planned for the MVP: User, Site, Sensor, SensorReading, VisionEvent, RiskAssessment, Incident, Alert and AuditLog.
- Field-level definitions and exact data types will be finalized after the first implementation pass and dataset structure are validated.
- The ERD represents the intended relationships between these entities and may be refined during development.

11.3 Proposed Schema Planning

- The proposed database schema will be version-controlled as development progresses.
- Database migrations will be introduced only if required by the implementation and will be documented alongside schema changes.
- Schema changes will be reviewed and tested in the development environment before being adopted in the project build.
- ERD is included as a technical appendix.

11.4 Privacy, Retention and Data Handling Principles

- Personal data: user identity and account information are minimized; worker identity is not required for core safety detection.
- Video privacy: raw camera streams are processed locally/ephemerally where possible; incident clips are optional and access-controlled.
- Biometric identification is explicitly out of scope.
- During development, only the data necessary for testing and demonstration will be retained, with retention settings defined according to the project environment.
- Backup and recovery will be considered as future deployment requirements; detailed RPO/RTO targets are outside the current synopsis scope.
- Planned controls: access to incidents and analytics will be role-controlled and audit logged.

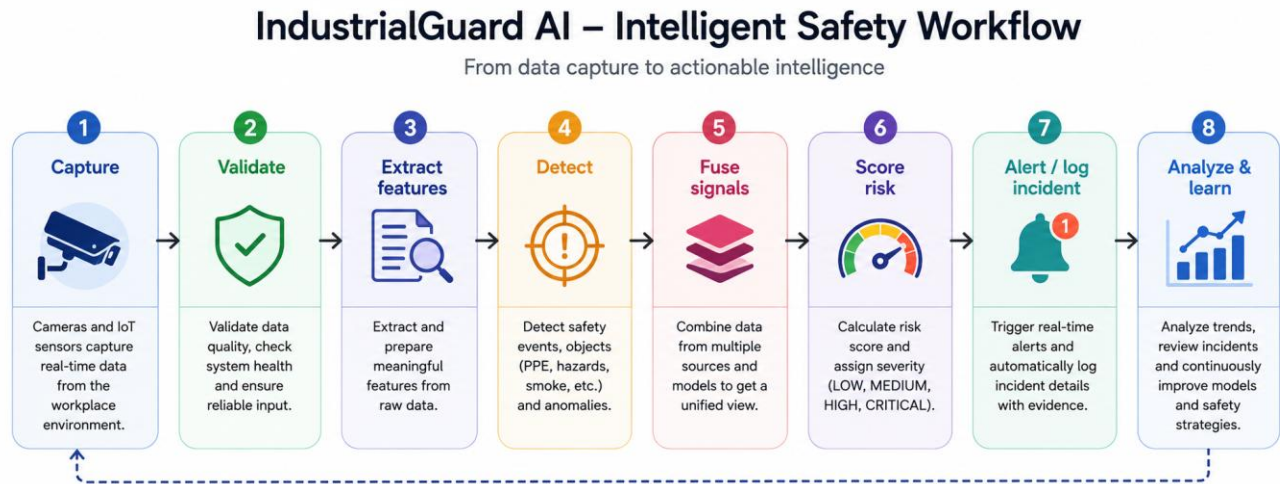
11.5 Proposed Data Lifecycle

IndustrialGuard AI – Workflow



The lifecycle is designed to minimise unnecessary retention of raw sensor/video data while preserving the metadata required for safety analytics and auditability.

12. Technical Workflow Diagrams



Key diagrams included in the design package

- State Transition Diagram: Sensor/vision event → risk state → incident state → resolution.
- Sequence Diagram: Device/CV source → API → ML/CV service → risk fusion → alert → dashboard.
- Use Case Diagram: Safety Officer, Shift Supervisor and Plant Manager interactions.
- Data Flow Diagram (DFD): Capture → Validate → Extract Features → Detect → Fuse Signals → Score Risk → Alert/Log Incident → Analyse/Learn.
- Entity Relationship Diagram (ERD): User, Site, Sensor, Reading, VisionEvent, RiskAssessment, Incident, Alert and AuditLog.
- System Architecture Diagram: End-to-end platform components and integration boundaries.

13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	Stable during controlled demonstration	Uptime monitor / logs
Latency	API p95	≤ 1000 ms	Application timing
Event response	Sensor/CV event → alert	≤ 3 s target under prototype conditions	Timestamp comparison
Error rate	5xx %	$\leq 5\%$ in controlled test runs	Backend logs
Security	Critical unresolved vulnerabilities	0 known critical issues before submission	Dependency / scanner review
Model quality	F1 score	Report and justify on held-out test set	ML evaluation notebook

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
ML	Offline evaluation	Python / Scikit-learn	Abhay	All test metrics reported	Model selected with evidence
Backend	Unit + integration	Pytest	Abhay	$\geq 70\%$ relevant code coverage	No P1/P2 defects
API	Integration	Postman	Anjali	All critical endpoints	All critical scenarios pass
UI	E2E	Playwright	Anjali	Core journeys	Pass rate $\geq 95\%$
System	End-to-end	Manual + scripted	Both	Core safety flow	Data → prediction → alert → incident works

13.3 Planned Development Environments

- Development → Controlled testing → Demonstration build
- Configuration switches may be used to evaluate experimental CV features without changing the core safety workflow.
- Representative and simulated datasets will be maintained for repeatable testing.

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Auth tokens	Token theft	Spoofing	High	Medium	HTTPS, short token TTL, secret rotation	Anjali
Sensor/API payloads	Tampering	Tampering	Medium	Medium	Schema validation, auth, audit logs	Anjali
Incident data	Unauthorised access	Information disclosure	High	Low	RBAC, least privilege, access logging	Abhay
Database	Injection	Tampering	High	Low	Parameterized queries / ORM validation	Anjali
Video evidence	Uncontrolled retention	Information disclosure	High	Medium	Minimal retention, encryption, access control	Both
Alerts	False or duplicate alerting	Repudiation / DoS	Medium	Medium	Rate limits, deduplication, event IDs	Both

14.2 AuthN/AuthZ

- Planned email/password authentication for prototype users
- Planned JWT-based authorization for API requests
- Planned role-based access control: Admin, Safety Officer, Supervisor
- Least-privilege access to incident evidence and configuration will be designed into the prototype.

14.3 Audit and logging

- The prototype is designed to log logins, configuration changes, risk decisions, incident updates and alert events.
- Never log passwords, raw secrets or unnecessary personal data.
- Use incident IDs and timestamps for traceability.

14.4 Compliance

- Academic project; follow institute policy and applicable data-protection requirements.
- No biometric identification or employment decisions are part of the prototype.
- Use only authorised/public/consented datasets and clearly label simulated data.

15. Development and Delivery Plan

15.1 Planned Development Milestones

- v0.1 - data ingestion + baseline model
- v0.2 - dashboard + prediction API
- v0.3 - CV event integration + risk fusion
- v0.4 - incidents + alerts + analytics
- Final milestone - end-to-end prototype, documentation and presentation

15.2 Development and Version Control Practices

- Planned quality workflow: lint → unit tests → build → controlled testing → demonstration build.
- Git branches, commits and pull requests will be used to review and track changes.
- Previous working versions of code and model artefacts will be retained so that development changes can be safely reverted when required.

15.3 Planned Operational Checks

Metric	Threshold	Alert to	Planned response approach
API p95 latency	> 1200 ms	Tech Lead	Inspect logs → query profile → optimise/revert/change review
Error rate	> 2%	Tech Lead	Inspect failing endpoint → reproduce → revert/change review/fix
Sensor heartbeat	Missing > 30 s	Safety Officer	Check device/connectivity → flag sensor offline
CV inference failure	Repeated failures	Tech Lead	Check model/runtime → disable CV flag safely → investigate

15.4 Planned Response Guidelines

- Backend issue: inspect logs → identify the failing component → reproduce in the development environment → correct and retest.
- Sensor issue: verify the simulated/device input → check connectivity → mark the source unavailable → continue testing with remaining valid signals.
- CV issue: verify the model/runtime → isolate the CV component during testing → preserve sensor monitoring → investigate before re-enabling the feature.
- Alert issue: inspect duplicate events → review threshold/cooldown logic → retest the triggering condition.

15.5 Communication plan

- Team standup: Mon/Wed/Fri or as agreed.
- Weekly status update to mentor.
- Bi-weekly demo of working increments.
- All scope changes recorded in a shared backlog and synopsis revision log.

16. Risks and Mitigations

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Schedule slip	Medium	High	12	Scope freeze; prioritise Must features; weekly demos	Both	Open
Limited real sensor data	High	Medium	12	Use simulated/academic datasets; document assumptions	Anjali	Open
CV model performance	Medium	High	12	Start with one PPE use case; report precision/recall; tune threshold	Anjali	Open
False alarms	Medium	High	12	Risk fusion; cooldowns; confidence thresholds; human review	Both	Open
Privacy concerns	Low	High	8	No biometrics; minimal retention; role access; anonymised evidence	Both	Open
API / integration failure	Medium	Medium	9	Mock interfaces; retries; validation; fallback sample data	Anjali	Open

17. Research and Evaluation

17.1 Study of Existing Platforms

Current commercial products confirm that computer vision and continuous site intelligence are already being applied to PPE compliance, near-miss detection, vehicle/pedestrian risk, area controls, reporting and operational workflows. Intenseye describes 50+ safety indicators and integrations with existing cameras and industrial systems; Protex AI combines cameras, operational data and safety analytics; Voxel uses existing camera infrastructure to surface risk and connect detection to workflows; these products demonstrate a mature market category rather than a purely academic concept.

17.2 Evaluation Using Representative Data

- Sensor model: create train/validation/test splits; compare baseline classifiers where practical; report precision, recall, F1 and confusion matrix.
- Vision model: use a held-out image/video set for helmet/vest detection; inspect false positives/false negatives and inference behaviour.
- System: test end-to-end timestamped flow from input → prediction → alert → incident.

17.3 Planned User/Stakeholder Feedback

Internal walkthroughs, mentor reviews and demonstration feedback will be used during development to refine the dashboard, alert wording, incident workflow and project scope. Any future pilot feedback will be treated as validation input rather than evidence of production safety performance.

17.4 KPI Tracking

- p95 API latency
- sensor-to-alert response time
- model precision/recall/F1
- CV detection precision/recall
- alert acknowledgement rate
- incident resolution time
- system uptime during demo

17.5 Research Direction

The research component will compare model choices, threshold strategies, risk-fusion methods and privacy-preserving deployment patterns. The report will distinguish measured results from proposed targets and from external vendor claims.

18. Appendices

18.1 Glossary

Term	Meaning
AI	Artificial Intelligence; methods that perform tasks requiring pattern recognition, prediction or decision support.
CV	Computer Vision; processing images/video to detect objects, activities or conditions.
PPE	Personal Protective Equipment such as helmets, safety vests and gloves.
ML	Machine Learning; algorithms that learn predictive patterns from data.
RBAC	Role-Based Access Control.
SLO	Service Level Objective; a target for system behaviour such as latency or uptime.
p95	95th percentile; a latency threshold below which 95% of measured requests fall.
Risk Fusion	Combining multiple signals into a single severity assessment.
EHS	Environment, Health and Safety management.

18.2 References

R1. Intenseye, Core AI: 50+ workplace safety indicators and computer-vision based EHS platform.

<https://www.intenseye.com/products/core-ai>

R2. Intenseye, Industrial Integrations: EHS AI, reporting/workflows, smart devices and IoT/OT integrations. <https://www.intenseye.com/products/industrial-integrations>

R3. Protex AI, Site Intelligence and Safety Overview. <https://www.protex.ai/> and <https://www.protex.ai/safety-overview>

R4. Protex AI, Privacy by design and on-site edge deployment. <https://www.protex.ai/guides/privacy>

R5. Voxel AI, Physical AI for industrial operations and safety. <https://www.voxelai.com/>

R6. Voxel AI, Operationalising AI for workplace safety. <https://www.voxelai.com/blog/ai-301-operationalizing-ai-for-workplace-safety>

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R8. GLA University synopsis template supplied by the department (2025-26).